Elections → negative to positive

Current → positive to negative (Ammeter) series

Electric Charge → axıyan axını zamanı keçən yüklərin sayı

$Q = I \cdot t$ 　 Q → charge (C)

I → current (A)

t → time (s)

Potential difference → Bir yükün 1 nöqtədən başqa nöqtəyə getməsi zamanı yaranan enerji fərqi

P. Difference → Volts (V) 　 $V = \frac{J}{C}$

$V = \frac{W}{Q}$ 　 W = work done (J) Joule 　 $W = N \cdot m$

Q = charge (C) coulomb 　 $J = \frac{kg \cdot m^2}{san^2}$

Electromotive force (e.m.f) → batareyanın və enerji mənbəyinin 1 yükə verdiyi enerji.

$\varepsilon = \frac{W}{Q}$ (V) 　 $V = \frac{J}{C}$

$\varepsilon = IR + Ir = I(R+r)$

lost volts $= Ir$

Series Circuit → components connected end to end

$I = const$

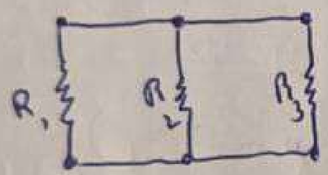
$R_{total} = R_1 + R_2 + R_3$

$V = I R$

- $V_{total} = V_1 + V_2 + V_3$

Paralel Circuit → components are connected accross each other's lead

$R_{total} = \frac{R_1 \cdot R_2}{R_1 + R_2}$



$V = const$.

$I_{total} = I_1 + I_2 + I_3$

Ohm law →

$I = \frac{V}{R}$ $\quad V = I R$

$R = Om$

Electricity → is movement of charge (electrons)

1) proton → + 3) neytron – 0 charge

2) electron → -

1) Electron move around the nuclers

2) Protons and neutrons stay in it.

+/+ or -/- repel (İtələyir)

+/- attract (yaxınlaşdırır)

Current → electric current is the flow of charge (axın)

unit → Amper (A) current dıması electron lora istəfə hərəkət edəcək, yayılacaq

Charges → ⊕ ⊖

When conductor is exposed to a potential difference, charge carries flow

Circuit → 1) electricity flows only in a complete circuit (closed loop)

2) electrons return to start point (+ → -)

If circuit is open → no flows

Components: 1) Energy source – battary

2) Conductor – wires path for current

3) Load → lamp

4) Switch → controls circuit

1) $V = IR$

$V = 9V$

$R = 3k\Omega = 3000\ \Omega$

$I = \frac{V}{R} = \frac{9}{3000} = 3 \cdot 10^{-3} = 3\ mA$

2) series → $R_{total} = 3 + 10 + 5 = 18 \cdot 10^{3}\ \Omega$

$V = 9V$ $I = \frac{9}{18 \cdot 10^{+3}} = 0{,}5 \cdot mA$

3) $U_1 = R_1 \cdot I = 3 \cdot 10^{3} \cdot 500 \cdot 10^{-6} = 1{,}5\ V$

$U_2 = R_2 \cdot I = 10 \cdot 10^{3} \cdot 500 \cdot 10^{-6} = 5\ V$ $1{,}5 + 5 + 2{,}5 = 9\ V$

$U_3 = R_3 \cdot I = 5 \cdot 10^{3} \cdot 500 \cdot 10^{-6} = 2{,}5\ V$

4) $\mathcal{E} = 9V$ $\mathcal{E} = I(R + r)$

$W = 1440$ $I = \frac{Q}{t} = \frac{240}{120} = 2\ A$ $R = \frac{V}{I} = \frac{6}{2} = 3\ \Omega$

$Q = 240$ $V = \frac{W}{Q} = \frac{1440}{240} = 6$

$t = 120\ s$

$9 = 2 \cdot (3 + r)$

$4{,}5 = 3 + r$

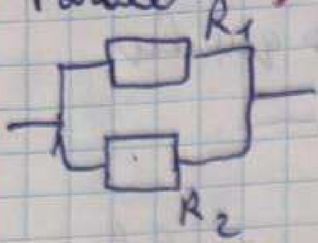
$r = 1{,}5$

5) $R = \frac{V}{I} =$

$V = \frac{W}{Q} = \frac{N \cdot m}{A \cdot s} = \frac{kg\ m}{s^{2} \cdot A \cdot s} =$

$= \frac{kg \cdot m}{A \cdot s^{+3}}$ $R = kg \cdot m \cdot A^{-2} \cdot s^{-3}$

Paralel !



$$R=\frac{R_1 \cdot R_2}{R_1+R_2}$$

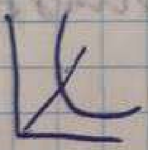
$$R=\frac{R}{n}$$

$U=const$

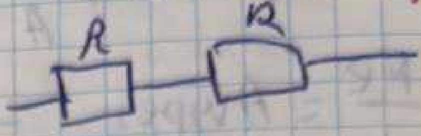
$q=q_1+q_2$

$$A=Q=\frac{U^2}{R}\cdot t$$

$$I=\frac{U}{R}$$



Ardıcıl !



$R=R_1+R_2$

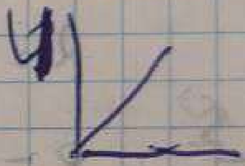
$R=nR$

$I=const$

$U=U_1+U_2$

$A=Q=I^2Rt$

$$I=\frac{U}{R}$$



Tam dövrə üçün Om !!

$$\varepsilon=\frac{A_k}{q}=V$$

Dövrə açıq olanda voltmetr EHQ

qapalı olanda voltmetr gərginlik